

ROBO CAR RACE (The Ultimate Track Challenge)

Event Overview

Robo Car Race – The Ultimate Track Challenge is an exhilarating test of design, precision, and control. In this competition, participants design and build their own remote-controlled robotic cars to navigate through a specially designed zig-zag track filled with challenging turns and time-critical sections. The event is all about speed, accuracy, and control — the robot that completes the track in the shortest possible time without penalties will be declared the winner. This competition combines innovation with performance, testing the participants’ technical skills, creativity, and teamwork under real racing conditions.

Eligibility Criteria

- Open to students from any college or department.
- Participation is strictly team-based (maximum 4 members per team).
- Prior registration is mandatory.

Participation Guidelines

- Teams must use self-made or prebuilt robotic cars.
- Teams must report to the event venue on time as per the schedule provided.
- Only approved tools and components can be used during the event.
- Participants must follow all safety and discipline protocols during testing and competition.

Design Specifications

Dimensions	Maximum length: 25 cm, width: 20 cm, height: 15 cm
Weight	Should not exceed 2 kg (including batteries)
Control Type	Wireless remote control (manual operation only)
Power Source	Onboard batteries only (no external power connection allowed)
Motors	DC motors or geared motors are allowed

Battery and Power

- Robots must use sealed, rechargeable batteries such as Li-ion, NiMH, or Ni-Cd.
- Maximum voltage allowed: 8V DC.
- Batteries must be securely mounted inside the robot chassis.
- Damaged or leaking batteries will lead to disqualification.
- No battery change is allowed during a race run.

Arena Description

- The race track will be zig-zag and plain, designed to test control and maneuverability.
- The surface will be smooth plywood or vinyl sheet, offering medium traction.

- Track width: 1 meter
- Length: Approximately 10 meters
- The track will include sharp turns, narrow lanes, and timing checkpoints.
- Obstacles such as minor humps or cones may be included to test stability.

Event Format

The competition will follow a multi-round elimination format:

1. Round 1 – Qualifier: Each team completes one timed run. Top performers qualify for the next round.
2. Round 2 – Semi-Finals: Qualified teams compete on a modified track with increased difficulty.
3. Round 3 – Final Challenge: The top teams race for the championship title and prizes.

Race Rules

- Each team will get two attempts in the qualifier round; the best time will be considered.
- Time will start when the car crosses the start line and stop when it crosses the finish line.
- Touching or resetting the robot during a run will add a penalty of 5 seconds per touch.
- Leaving the track or skipping sections will lead to disqualification for that run.
- Robots must complete the track without external assistance once the run begins.
- Judges' decisions will be final and binding in all matters.

Judges' and Organizers' Rights

- The organizers reserve the right to change rules or track layout if necessary.
- In case of any dispute, the judges' decision will be final.
- Any form of misconduct or tampering will result in disqualification.